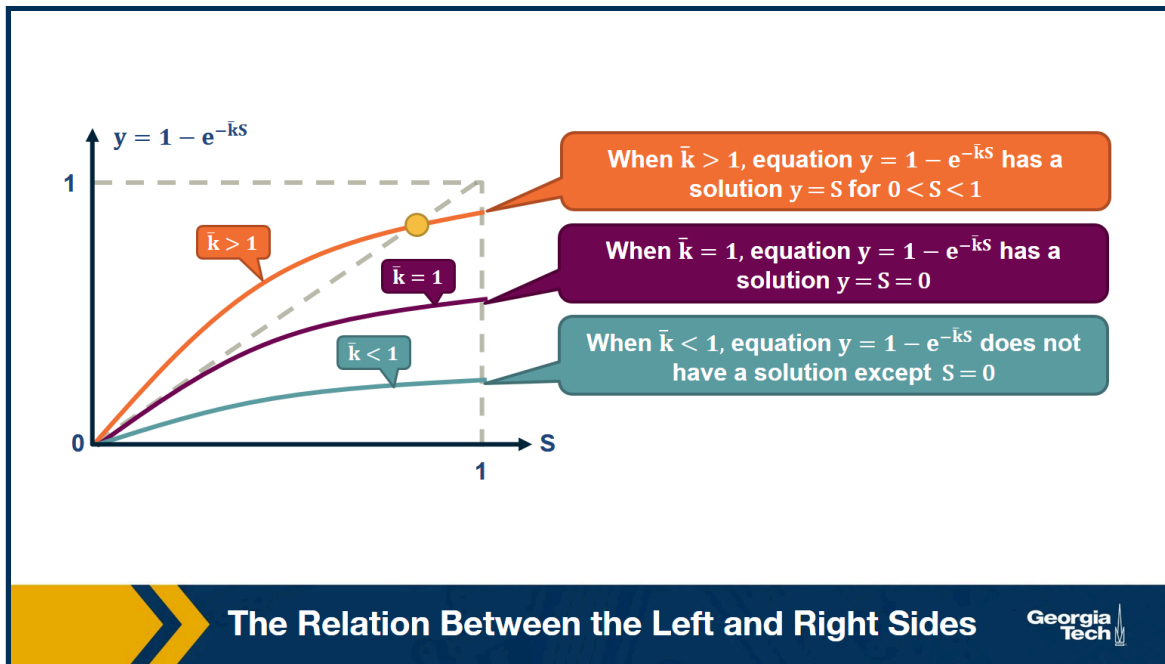


## L3: Size of LCC in $G(n,p)$ as Function of $p$

We can derive the relation between  $p$  and the size of the LCC as follows:



Suppose that  $S$  is the probability that a node belongs in the LCC. Another way to think of  $S$  is as the expected value of the fraction of network nodes that belong in the LCC.

Then,  $\bar{S} = 1 - S$  is the probability that a node does NOT belong in the LCC.

That probability can be written as

$$\bar{S} = \left( (1-p) + p \cdot \bar{S} \right)^{n-1}$$

The first term refers to the case that a node  $v$  is not connected to another node, while the second term refers to the case that  $v$  is connected to another node that is **not** in the LCC.

Since,  $p = \frac{\bar{k}}{n-1}$ , the last equation can be written as:

$$\bar{S} = \left( 1 - \frac{\bar{k}}{n-1} (1 - \bar{S}) \right)^{n-1}$$

$$\ln \bar{S} = (n-1) \ln \left( 1 - \frac{\bar{k}}{n-1} (1 - \bar{S}) \right) \approx - (n-1) \frac{\bar{k}}{n-1} (1 - \bar{S})$$

$$S = 1 - e^{-\bar{k}S}$$

The visualization shows the relation between the left and right sides of the previous equation, i.e., the relation between  $S$  and  $1 - e^{-\bar{k}S}$ .

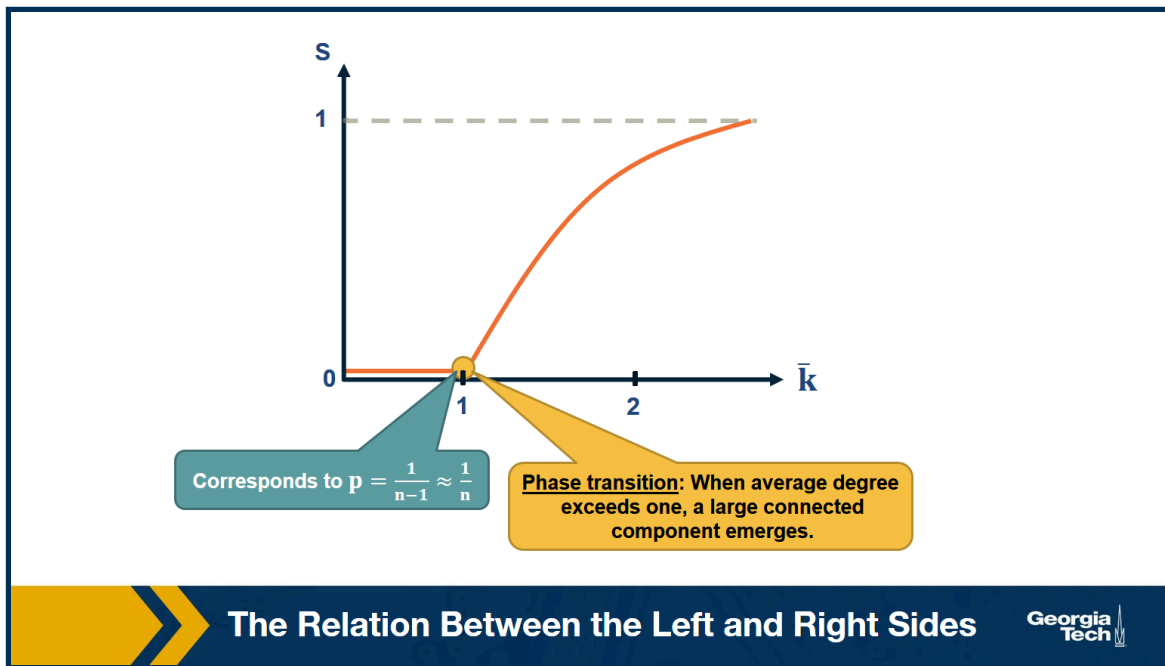
The equality is true when the function  $y = 1 - e^{-\bar{k}S}$  crosses the diagonal  $x=y$  line.

Note that the derivative of  $y$  with respect to  $S$  is approximately  $\bar{k}$  when  $S$  approaches 0.

So, if the average degree is larger than one, the function  $y(S)$  starts above the diagonal. It has to cross the diagonal at a positive value of  $S$  because its second derivative of  $y(S)$  is negative. That crossing point is the solution of the equation

**$S = 1 - e^{-\bar{k}S}$ . This means that if the average degree is larger than one ( $\bar{k} > 1$ ), the size of the LCC is  $S > 0$ .**

On the other hand, if the average node degree is less (or equal) than 1, the function  $y(S)$  starts with a slope that is less (or equal) than 1, and it remains below the diagonal  $y=x$  for positive  $S$ . This means that if the average node degree is less or equal than one, the average size of the LCC in a  $G(n,p)$  network includes almost zero nodes.



The visualization shows how  $S$  increases with the average node degree  $\bar{k}$ . Note how the LCC suddenly “**explodes**” when the average node degree is larger than 1. This is referred to as “phase transition”. A phase transition that we are all familiar with is what happens to water when its temperature reaches the freezing or boiling temperature: the macroscopic state changes abruptly from liquid to solid or gas. Something similar happens with  $G(n,p)$  when the average node degree exceeds the critical value  $\bar{k} = 1$ : the network suddenly acquires a “giant connected component” that includes a large fraction of all network nodes.

Note that the critical point corresponds to a connection probability of  $p = \frac{1}{n-1} \approx \frac{1}{n}$ , because  $\bar{k} = (n - 1) * p$ .